

```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv("dataset.csv")
```

```
In [3]: data
```

```
Out[3]:
```

	Year	Average_Temperature_C	Annual_Rainfall_mm	Extreme_Weather_Eve
0	2000	25.25	1278.46	
1	2001	25.41	1256.74	
2	2002	25.55	1221.68	
3	2003	25.64	1203.52	
4	2004	25.71	1189.65	
5	2005	25.92	1152.39	
6	2006	26.05	1138.42	
7	2007	26.24	1115.76	
8	2008	26.31	1089.24	
9	2009	26.48	1067.58	
10	2010	26.55	1043.27	
11	2011	26.72	1023.65	
12	2012	26.89	1001.23	
13	2013	27.02	985.47	
14	2014	27.18	973.65	
15	2015	27.32	958.48	
16	2016	27.45	943.76	
17	2017	27.59	929.28	
18	2018	27.73	915.42	
19	2019	27.86	904.38	
20	2020	27.98	894.26	
21	2021	28.12	883.14	
22	2022	28.24	872.53	
23	2023	28.36	865.32	
24	2024	28.50	859.74	

```
In [4]: data.shape
```

```
Out[4]: (25, 7)
```

```
In [5]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25 entries, 0 to 24
Data columns (total 7 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   Year                                  25 non-null     int64
 1   Average_Temperature_C                25 non-null     float64
 2   Annual_Rainfall_mm                   25 non-null     float64
 3   Extreme_Weather_Events               25 non-null     int64
 4   Soil_Moisture_Percentage              25 non-null     float64
 5   Crop_Yield_ton_per_hectare           25 non-null     float64
 6   Unnamed: 6                           0 non-null      float64
dtypes: float64(5), int64(2)
memory usage: 1.5 KB
```

```
In [6]: data.isnull().sum()
```

```
Out[6]: Year                                0
Average_Temperature_C                      0
Annual_Rainfall_mm                         0
Extreme_Weather_Events                     0
Soil_Moisture_Percentage                    0
Crop_Yield_ton_per_hectare                  0
Unnamed: 6                                25
dtype: int64
```

```
In [7]: data.describe()
```

```
Out[7]:
```

	Year	Average_Temperature_C	Annual_Rainfall_mm	Extreme_We
count	25.000000	25.000000	25.000000	
mean	2012.000000	26.882800	1030.680800	
std	7.359801	1.007466	132.788561	
min	2000.000000	25.250000	859.740000	
25%	2006.000000	26.050000	915.420000	
50%	2012.000000	26.890000	1001.230000	
75%	2018.000000	27.730000	1138.420000	
max	2024.000000	28.500000	1278.460000	

```
In [8]: data.columns
```

```
Out[8]: Index(['Year', 'Average_Temperature_C', 'Annual_Rainfall_mm',
              'Extreme_Weather_Events', 'Soil_Moisture_Percentage',
              'Crop_Yield_ton_per_hectare', 'Unnamed: 6'],
              dtype='object')
```

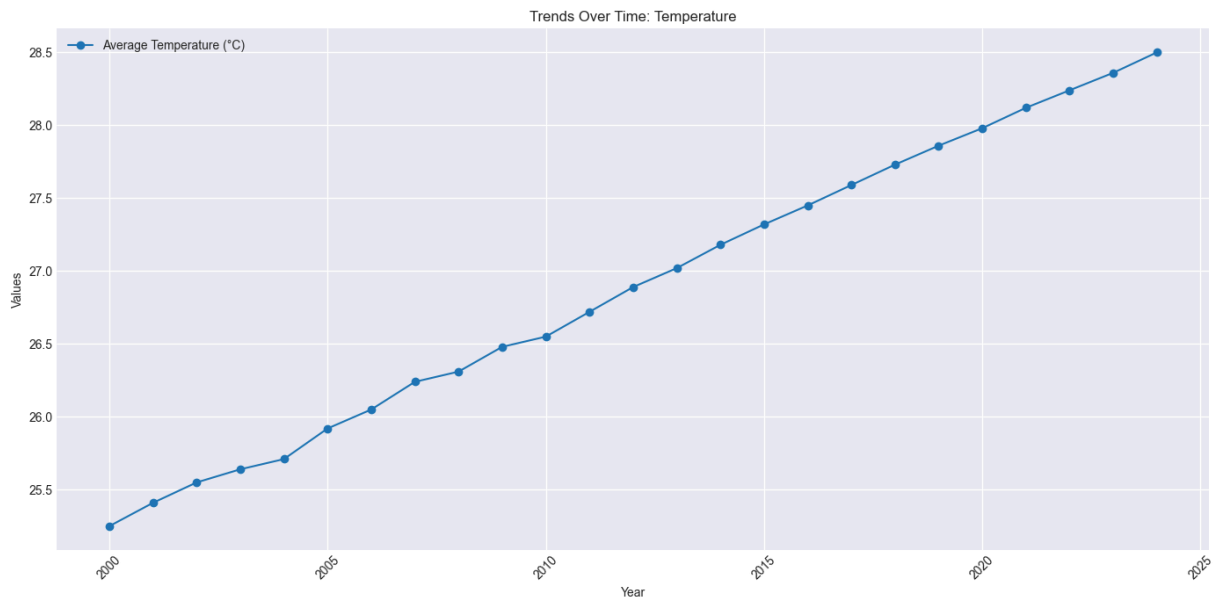
```
In [9]: plt.style.use('seaborn-v0_8-darkgrid') # Updated style name for compatibility

# Create line plots for temperature, rainfall, and crop yield over time
plt.figure(figsize=(14, 7))

# Average Temperature
plt.plot(data['Year'], data['Average_Temperature_C'], label='Average Temperature')

# Add labels and title
plt.title('Trends Over Time: Temperature')
plt.xlabel('Year')
plt.ylabel('Values')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

# Show plot
plt.show()
```



```
In [10]: plt.style.use('seaborn-v0_8-darkgrid') # Updated style name for compatibility

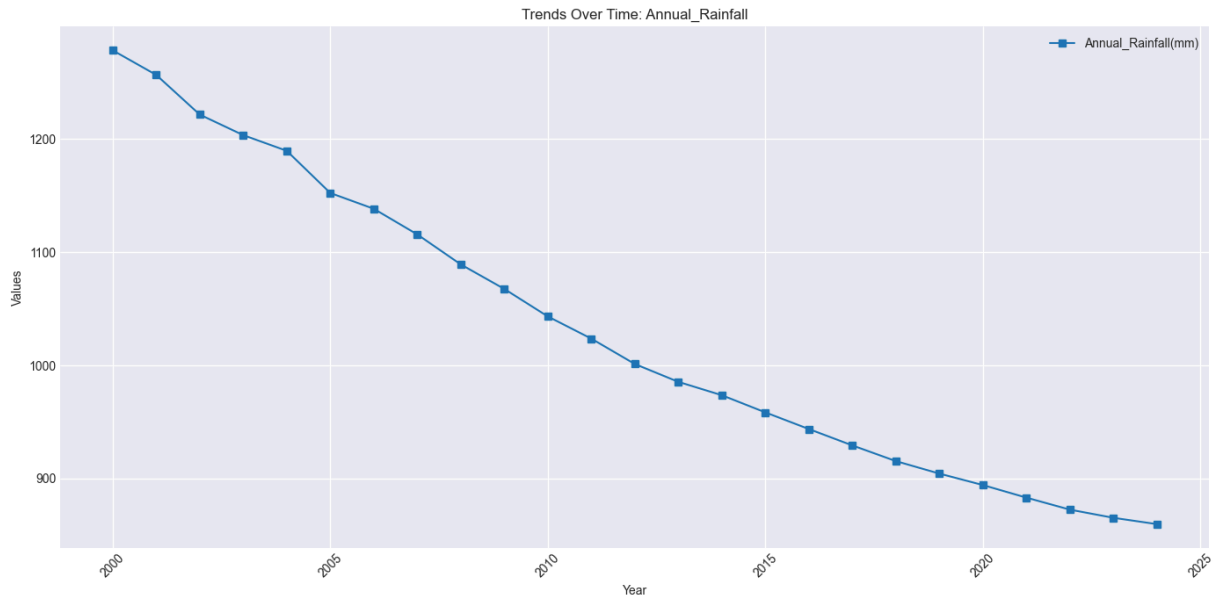
# Create line plots for temperature, rainfall, and crop yield over time
plt.figure(figsize=(14, 7))

# Average Temperature
plt.plot(data['Year'], data['Annual_Rainfall_mm'], label='Annual Rainfall(mm)')

# Add labels and title
plt.title('Trends Over Time: Annual_Rainfall')
plt.xlabel('Year')
plt.ylabel('Values')
plt.legend()
```

```
plt.xticks(rotation=45)
plt.tight_layout()

# Show plot
plt.show()
```



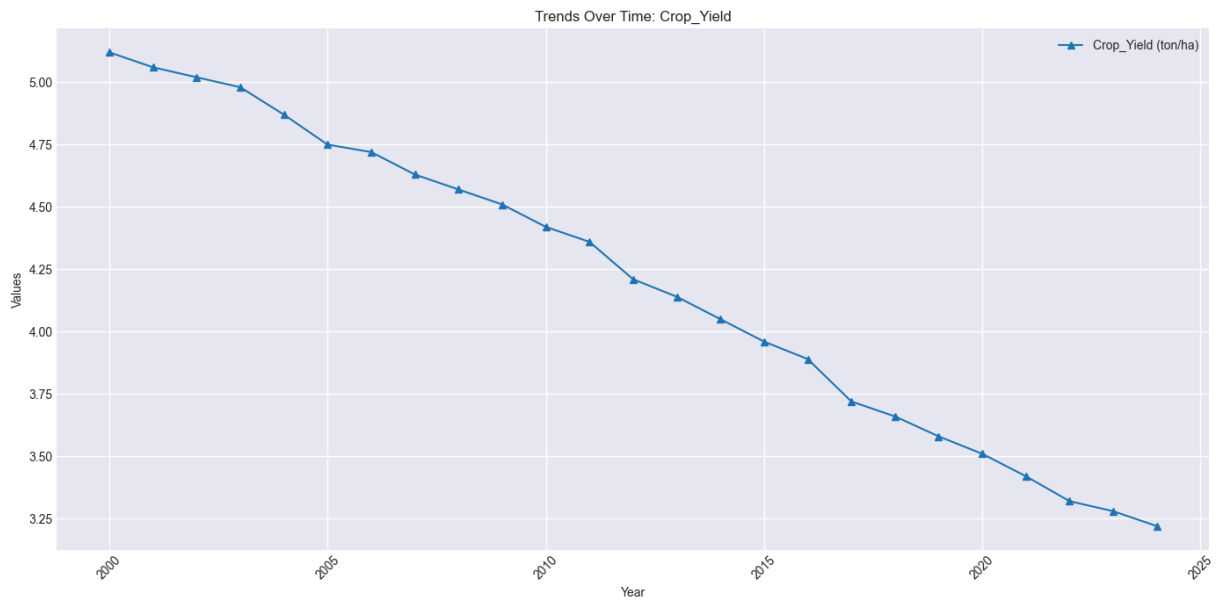
```
In [11]: plt.style.use('seaborn-v0_8-darkgrid') # Updated style name for compatibility

# Create line plots for temperature, rainfall, and crop yield over time
plt.figure(figsize=(14, 7))

# Average Temperature
plt.plot(data['Year'], data['Crop_Yield_ton_per_hectare'], label='Crop_Yield')

# Add labels and title
plt.title('Trends Over Time: Crop_Yield')
plt.xlabel('Year')
plt.ylabel('Values')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

# Show plot
plt.show()
```



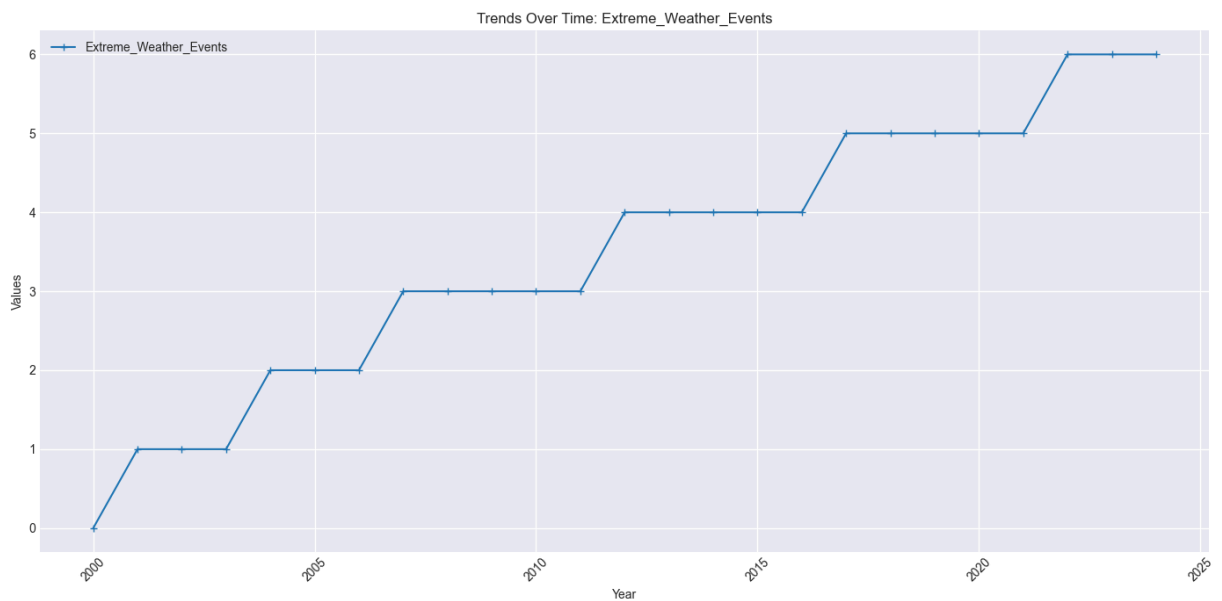
```
In [12]: plt.style.use('seaborn-v0_8-darkgrid') # Updated style name for compatibility

# Create line plots for temperature, rainfall, and crop yield over time
plt.figure(figsize=(14, 7))

# Average Temperature
plt.plot(data['Year'], data['Extreme_Weather_Events'], label='Extreme_Weather_Events')

# Add labels and title
plt.title('Trends Over Time: Extreme_Weather_Events')
plt.xlabel('Year')
plt.ylabel('Values')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

# Show plot
plt.show()
```



```
In [13]: plt.style.use('seaborn-v0_8-darkgrid') # Updated style name for compatibility

# Create line plots for temperature, rainfall, and crop yield over time
plt.figure(figsize=(14, 7))

# Average Temperature
plt.plot(data['Year'], data['Soil_Moisture_Percentage'], label='Soil_Moisture_Percentage')

# Add labels and title
plt.title('Trends Over Time: Soil_Moisture_Percentage')
plt.xlabel('Year')
plt.ylabel('Values')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

# Show plot
plt.show()
```

